Questions and Answers

Part 2.1

Question 1

The theoretical phase shift between a sinewave and a cosine wave is 90°. If your measurement isn't exactly 90°, what could explain the difference?

Answer:
Real-world components introduce parasitic capacitance and inductance, which cause unintended phase lag. In addition, the Master Signals module derives the cosine wave electronically from the sine wave, and small component tolerances in the phase-shifting network result in deviations from the ideal 90° phase shift.

Question 2

Calculate the Buffer module's gain (on its present gain setting).

Answer:

$$A_v = \frac{V_{out}}{V_{in}} = \frac{4.11}{3.92} = 1.05$$

Question 3

Calculate the Buffer module's new gain.

Answer:

$$A_v = \frac{V_{out}}{V_{in}} = \frac{360mV}{3.92V} = 0.09V$$

Question 4

In terms of the gain figure, what's the difference between gain and attenuation?

Answer:
A gain greater than 1 indicates amplification. A gain between 0 and 1 indicates attenuation. A negative gain indicates a 180° phase inversion.

Question 5

What do you think the output signal would look like if the buffer's gain was sufficiently large?

Answer:
The signal would clip due to the voltage supply limits, flattening the peaks of the waveform and eventually resembling a square wave.

Question 6

Why is the Master Signals module's 100 kHz SINE output inaudible?

Answer:
Human hearing is limited to approximately 20 Hz–20 kHz. A 100 kHz signal is ultrasonic and therefore inaudible.

Part 2.2

Question 1

What aspect of the Adder module's performance does the G control vary?

Answer:
It varies the gain (weighting) applied to input A before summation.

Question 2

What is the range of gains for the A input?

Answer:
The gain typically ranges from 0 to approximately 2.

Question 3

Compare the results in Tables 1 and 2. What can you say about the Adder module's two inputs in terms of their gain?

Answer:
Both inputs are identical and independent, providing matched gain performance.

Question 4

What is the relationship between the amplitude of the signals on the Adder module's inputs and output?

Answer:

$$V_{OUT} = (G\ x\ V_A) + (G\ x\ V_B)$$

Question 5

This module's output signal can be phase shifted by different amounts:

- ☐ but it always leads the input signal
- ☐ but it always lags the input signal
- ☑ and can either lead or lag the input signal

Question 6

Is the Phase Shifter module capable of shifting a signal by 360°?

Answer:
No. A single phase shifter is typically limited to about 180°. Achieving 360° requires cascading multiple phase shifters.

Part 2.3

Question 1

What happens to the Variable DCV module's VDC output when you turn the VDC control clockwise?

Answer:
The DC output voltage increases.

Question 2

What happens to the frequency of the VCO module's output when you turn the VDC control clockwise?

Answer:
The output frequency increases.

Question 3

What happens to the Variable DCV module's VDC output when you turn the VDC control anti-clockwise?

Answer:
The DC voltage decreases.

Question 4

What happens to the frequency of the VCO module's output when you turn the VDC control anti-clockwise?

Answer:
The output frequency decreases.

Part 3

Question 1

Is the Adder module's measured output voltage exactly 8 Vp-p as theoretically predicted?

Answer:
No.

Question 2

What are two reasons for this?

Answer:
Component tolerances and loading effects from the measurement equipment.

Question 3

What are two reasons for the output not being 0 V as theoretically predicted?

Answer:
Imperfect phase alignment and slight gain mismatches prevent perfect cancellation.

Question 5

What can be said about the phase shift between the signals on the Adder module's two inputs now?

Answer:
They are in phase (0° phase difference).

Question 6

What can be said about the gain of the Adder module's two inputs now?

Answer:
Both gains are set to maximum to achieve the largest output.

Part 4

Question 1

In what way is the Adder module's output now different to the signal out of the Master Signals module's 2k/2 SINE output?

Answer:
The output contains a DC offset, shifting the sine wave above 0 V.

Question 2

What feature of the Multiplier module's output suggests that it's an AM signal?

Answer:
The presence of a symmetrical envelope around the carrier.

Question 3

The AM signal is a complex waveform consisting of more than one signal. Is one of the signals a 2 kHz sinewave? Explain your answer.

Answer:
Yes. The 2 kHz sinewave is the modulating (message) signal.

Question 4

For the given inputs to the Multiplier module, how many sinewaves does the AM signal consist of, and what are their frequencies?

Answer:
Three components:

- Carrier at 100 kHz
- Upper sideband at 102 kHz
- Lower sideband at 98 kHz

Question 5

Why is there still a signal out of the Multiplier module even when you're not talking, whistling, etc.?

Answer:
The carrier is always present, and internal noise or bias in the modulating input still causes modulation.

Question 6

What is the relationship between the message's amplitude and the amount of the carrier's modulation?

Answer:
They are directly proportional.

Question 7

What is the problem with the AM signal when it is over-modulated?

Answer:
Over-modulation causes envelope distortion and phase reversal, preventing proper demodulation.

Question 8

What do you think is a carrier's maximum modulation index without over-modulation?

- ☐ A minus number
- ☐ 0
- ☑ 1
- ☐ Greater than 1